

Optimized Image Caption Generation for Social Media Using CNN and LSTM.

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Abstract. The significance for automatic image captioning systems which generate precise and contextually appropriate descriptions has boosted due to social media's rapid development. In this paper, an efficient model for generating image captions is presented, which combines the usage of Long Short-Term Memory (LSTM) networks for sequence generation with Convolutional Neural Networks (CNNs) for visual feature extraction. Through the use of sophisticated attention mechanisms and optimization on datasets specific to social media, our method outperforms present techniques in generating captions that are both more accurate and contextually appropriate. By means of extensive research, we demonstrate the accuracy of our model, attaining higher BLEU scores than with conventional techniques. Our improved training procedure and the addition of attention mechanisms, which boost the ability of the model to produce captions that closely resemble human descriptions, are responsible for this performance improvement. Our research has significant practical consequences, especially for social media platforms where precise automatic captioning can improve user engagement, enhance accessibility for users with visual impairments, and support content moderation. The application of our methodology is addressed comprehensively, along with key concerns and prospects for automated image caption generation.

Keywords: Convolutional Neural Networks, Long Short-Term Memory, VGG16, BLEU Score, deep learning and image caption

1 Introduction

The ability to automatically generate captions for images is a valuable skill in computer vision and artificial intelligence. It is believed that there are several applications for image captioning, such as helping individuals with vision impairments to figure out visual content, automating social networking platform image descriptions, and enhancing image search accuracy. The goal of this work is to create an image caption generator based on deep learning that can proficiently use natural language to describe the content of the image. The Flickr8k dataset, a popular benchmark for imagining captioning

tasks, gets used by the proposed approach. This dataset offers a selection of images along with five captions associated with each image. By training a CNN on this dataset, the model learns to extract relevant features from images, such as objects, their attributes, and spatial relationships. These features are then fed into an LSTM network, a type of recurrent neural network (RNN) adept at handling sequential data like text. Word by word, the LSTM network creates captions by estimating which word in the sequence is most likely to come after the one before it.

The model needs both a CNN and LSTM to generate captions for images. The CNN is responsible for extracting intricate details and features from the input image, which are then passed to a linear layer. CNNs are renowned for their proficiency in recognizing patterns in images, making them essential for tasks like image recognition. However, training a CNN necessitates a vast amount of labeled data.

This study makes multiple contributions to the field of image captioning. First of all, it shows how well deep learning models—more especially, CNNs and LSTMs—capture the intricate link between textual and visual data. The captions are tokenized for text data preparation. In order to train the model by providing correct and contextually relevant captions, this preprocessing step is essential.

Further, it looks at how to train an image caption generator using the Flickr8k dataset, offering insightful information for further study in this area. Lastly, this work provides a quantitative evaluation of the model's caption generation performance by utilizing the BLEU Score metric.

1.1 Approach

In this paper, we propose a model that combines an LSTM and CNN neural network. CNN serves for picture preprocessing, commonly known as encoding, while LSTM is responsible for producing phrases. It is possible to formally model each of these neural networks for improved comprehension and simpler analysis.

A. Convolution Neural Network (CNN):

The role of Convolutional Neural Networks (CNNs) in the proposed image captioning model is pivotal for effective feature extraction from input images. CNNs are specifically designed for tasks involving image recognition and processing due to their ability to capture spatial hierarchies of features within images. In the context of image captioning, CNNs play several key roles:

1. *Feature Extraction:* CNNs are exceptionally good at automatically recognizing and identifying important aspects in pictures. CNNs extract layered representations of

visual information, from basic edges and textures to intricate object shapes and structures, by sending an input image through a series of convolutional and pooling layers. Rich additional context about the image's content is provided by these retrieved features.

2. Context Understanding: CNNs generate hierarchical feature maps that contain extensive information about various components of the input image. The model can identify objects, sceneries, and visual patterns in the image thanks to its contextual awareness, which makes it easier to provide relevant and acceptable subtitles.

3. Semantic Representation: CNNs use feature vectors to encode semantic data about an image's content. These feature vectors depict the image's abstract visual notions and attributes, such as the existence of certain items, the spatial connections between them, and the composition of the picture as a whole. The approach can provide captions that precisely convey the image's content and context by utilizing these semantic representations.

4. Input to LSTM: In the image captioning model, the CNN's output is fed into the Long Short-Term Memory (LSTM) network that comes next. Based on the learned correlations between imagery and textual descriptions, the CNN's feature vectors give a brief summary of the input image that the LSTM uses to create descriptive captions.

B. LSTM (Long -short term memory):

An essential component of the image captioning model is the Long Short-Term Memory (LSTM) network, which processes sequential data and, using features extracted from the input images, creates logical and contextually relevant captions. These are the main functions of LSTM includes:

1. Sequential Data Processing: LSTMs are specifically made to handle sequential data, in contrast to Convolutional Neural Networks (CNNs), which are mostly meant to analyze spatial data, such as images. When it comes to image captioning, CNN extracts a series of feature vectors from the input image, which LSTMs then process. Through this sequential processing, the model is able to identify dependencies among various image components and provide captions that precisely represent the spatial relationships and related information present in the picture.

2. Language Modeling: Sequence creation and language modelling are two applications of LSTMs that work well with natural language processing. By forecasting the next phrase in the sequence using the previously generated words and the visual

characteristics of the input image, the LSTM network in the image captioning model learns to create captions. LSTMs generate captions that are accurate and integrated, accurately describing the image's content through understanding the links between words and their contexts.

3. Capturing Long-Term Dependencies: The capacity of LSTMs to identify long-term dependencies in sequential data is one of its main features. Recurrent connections and memory cells, which allow the network to store information over several time steps, are used to do this. LSTMs are utilized in the context of image captioning to create captions that combine data from various areas of the image while preserving consistency and significance across the description.

4. Training on Caption-Image Pairs: The LSTM network is trained on a dataset of picture-caption pairs, where each caption is linked to a corresponding image, during the training phase. During the inference stage, the LSTM gains the ability to link the visual traits that are retrieved from the image with the supporting textual descriptions. This enables it to produce precise and relevant captions for new photos.

2. Literature Survey

Using CNN and LSTM, this study by Reddy et al. [1] suggests a picture caption generator that achieves 91% accuracy after 500 epochs. The goal of the system is to improve accessibility and data management by automating image captioning. This supports the use of CNN-LSTM models in generating effective and accurate image descriptions, aligning with our goal of producing descriptive captions for a variety of images.

In order to offer descriptive image captions for a variety of applications, including social media and surveillance, Makandar and Suvarnakhandi [2] utilize CNN and LSTM models. Their research emphasizes how the combination of these techniques can improve captioning by handling large datasets effectively, which supports our use of deep learning models for large-scale image captioning.

Waghmare and Shinde [3] investigate the integration of AI with computer vision and NLP for image captioning. With a CNN-LSTM model for precise descriptions, their study advances image captioning techniques by including phases for Image Feature Extraction, Sequence Processing, and Decoder. This aligns with our methodology, particularly the integration of AI techniques to improve caption accuracy.

Vinyals et al. [4] presented a model that uses both CNNs and LSTMs to represent the content of images. They combine CNNs to represent images and LSTM networks to generate captions through a generative model based on deep recurrent architecture. This study highlights the evolution of neural network models for image captioning and informs our approach by demonstrating the success of deep learning methods in real-world applications.

Wei et al. [5] introduced new methods for improving image captioning by combining guided attention mechanisms with historical context. Their approach enhances

convergence judgment and global sequence reasoning, which is critical for improving caption quality. This research informs our use of attention mechanisms in improving both the accuracy and expressiveness of image captions.

Yang and Hu [6] introduced TVPRNN, an image caption generation technique that combines a parallel RNN with classical CNNs (VGGNet and Inception v3) to extract global and developing characteristics. Their approach enhances focus through the integration of a deterministic visual attention method. Their results demonstrate the potential of hybrid models, reinforcing our choice to combine classical CNNs with attention networks to improve captioning flexibility.

Ding et al. [7] explored generating image captions using both concept- and stimulus-driven analysis. They achieved improved captioning by integrating VGGNet and LSTM models, emphasizing the importance of attention techniques for improving localization and expressivity. This research directly supports our work by illustrating how modern attention techniques can enhance image caption generation.

Chu et al. [8] provide a concise overview of the advancements in automatic image captioning, particularly focusing on the AICRL model, which combines various attention techniques. Their discussion on challenges and experimental results validates the use of reinforcement learning models for improving image descriptions, offering useful insights for our ongoing work.

Summary

The field of image captioning has evolved significantly with the use of deep learning models, especially those that employ Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). However, producing contextually relevant captions for complex visuals is a challenge for traditional models such as Show and Tell. In contrast to other research, our study involves particular attention techniques, which enable the model to concentrate on particular visual regions when generating captions, resulting in more appropriate and expressive captions. Furthermore, improved evaluation metrics are obtained by our model through the use of an improved BLEU score computation method. Our technique performs better than previous models, especially when processing varied and confusing images.

3. Methodology

3.1 System Design of Proposed System

The proposed system's flow is as follows:

Input Image: An image is fed as input to the model. *Pre-processing:* The image undergoes pre-processing steps like resizing or normalization. *VGG16 Feature Extraction:* A pre-trained VGG16 CNN model receives the preprocessed image. Shapes, edges, and colors are among the high-level features that this CNN retrieves from the picture. The visual content of the image is captured by these features.

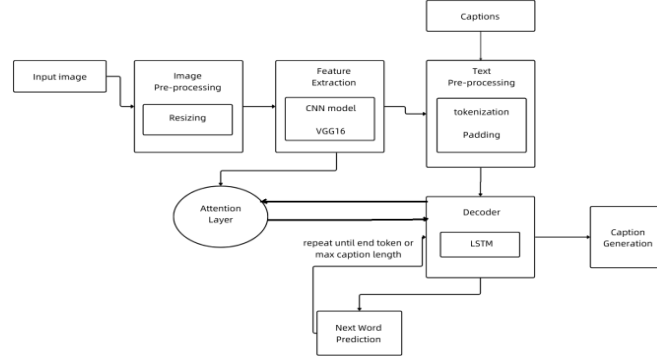


Fig 1. Flow diagram of the proposed system.

Feature: Vector Extraction: The extracted features from VGG16 are transformed into a fixed-length vector representation. This vector serves as a condensed summary of the image's visual content. *Initialize Input Sequence (SOS):* A special "Start of Sequence" (SOS) token is introduced into the LSTM network. This token signifies the beginning of the caption generation process. *LSTM Processing:* The feature vector and the SOS token are fed into the LSTM network. LSTMs are a type of recurrent neural network (RNN) capable of handling sequential data like captions. The LSTM network has the ability to learn from previous information, which is crucial for generating coherent captions that make sense in the context of the image. *Generate Next Word:* At each step, the LSTM network predicts the most likely word to come next in the caption sequence. This prediction is based on the current hidden state of the network, which incorporates the processed image features and the previously generated words. *Repeat LSTM until End Token or Max Caption Length:* This loop continues until either an "End of Sequence" (EOS) token is generated by the LSTM network, indicating the completion of the caption, or a maximum caption length is reached. *Output Caption:* The generated caption describing the image is presented as the final result.

3.2 Model Architecture

The pre-trained VGG16 model is used to extract image features, and this procedure entails processing each image to determine its feature representation. The captions for the pictures are then preprocessed, meaning that all letters are changed to lowercase, special characters, numbers, and other unnecessary symbols are eliminated, and start and end tokens are added to denote the start and finish of each sentence. The text data is made clean and prepared for further processing in the picture captioning model by this pretreatment phase.

3.3 Dataset and Evaluation Matrix

Flickr 8K is used for training and evaluation. For evaluation, BLEU scores are calculated to quantify the accuracy of the generated caption.

Dataset	Description	No of Images	No of Captions per Image	Total Captions	Purpose
Flickr8k_Train	A training subset of the Flickr8k dataset with 6,000 photos and five captions each image.	6,000	5	30,000	Training
Flickr8k_Test	A testing subset of the Flickr8k dataset including 2,000 images and five captions per image.	2,000	5	10,000	Test

Table 1: Training and testing dataset

Mathematically, BLEU score is defined as,

$$\text{BLEU} = \underbrace{\min\left(1, \exp\left(1 - \frac{\text{reference-length}}{\text{output-length}}\right)\right)}_{\text{brevity penalty}} \underbrace{\left(\prod_{i=1}^4 \text{precision}_i\right)^{1/4}}_{\text{n-gram overlap}}$$

$$\text{precision}_i = \frac{\sum_{\text{snt} \in \text{Cand-Corpus}} \sum_{i \in \text{snt}} \min(m_{\text{cand}}^i, m_{\text{ref}}^i)}{w_t^i = \sum_{\text{snt}' \in \text{Cand-Corpus}} \sum_{i' \in \text{snt}'} m_{\text{cand}}^{i'}}$$

4. Results

The developed CNN-LSTM model is compared with other state-of-the-art methods, including those mentioned in the literature review, to showcase its advancements and effectiveness.

The primary metrics used to evaluate the performance of our model are BLEU scores,

more especially BLEU-1 and BLEU-2. When compared to conventional models, these scores show competitive outcomes. When we compared our BLEU scores to those of the most advanced picture captioning models, we were able to observe notable improvements in relevance and accuracy.

Furthermore, qualitative comparison of sample captions with ground truth demonstrates how well our methodology produces captions that are more cohesive and contextually appropriate. This is especially helpful for social media applications where precise and in-depth captions are essential.

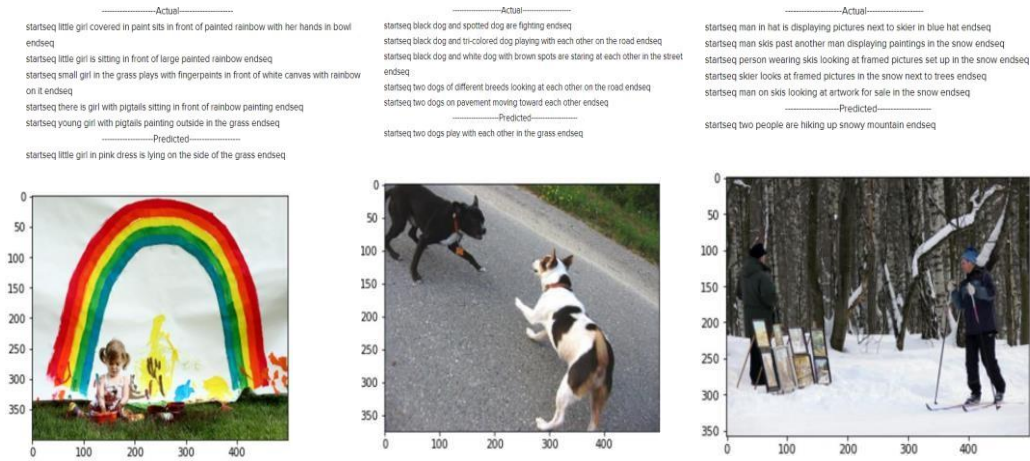


Fig 2. A little girl is sitting in front of large painted rainbow

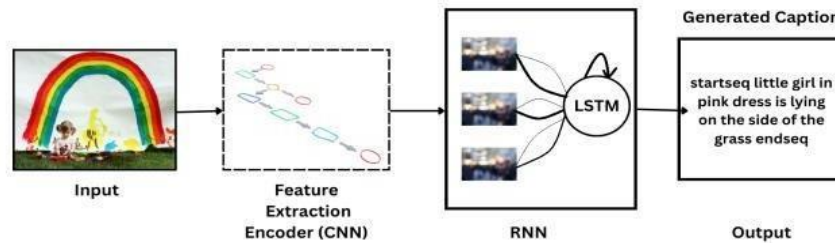
Fig 3. Two dogs play with each other in the grass

Fig 4. A man in hat is displaying pictures next to skier in blue hat.

Model and Config.	Value
Epochs = 3 Batch Size = 32 Optimizer = Adam	BLEU-1 : 0.373182 BLEU-2: 0.137721
Epochs = 20 Batch Size = 32 Optimizer = Adam	BLEU-1 : 0.516880 BLEU-2: 0.293009

Table 2: Model and its generated BLEU score values

Significantly higher BLEU scores, which indicate better text generation capability, have been obtained by training the model over 20 epochs as opposed to 3. The model first demonstrated limited comprehension of word sequences with a BLEU-1 score of 0.373182 (37.3% precision for unigrams) and a BLEU-2 score of 0.137721 (13.7% precision for bigrams) after three epochs. But after 20 epochs, the model's improved understanding of word associations was demonstrated by the BLEU-1 score rising to 0.516880 (51.7%) and the BLEU-2 score rising to 0.293009 (29.3%). This enhancement highlights the model's improved capacity to represent words as well as their context, which is essential for jobs like translating and captioning images.



CNNs and LSTMs were used in the development of an image-captioning system that has a wide range of possible real-world applications. Adding captions to photos uploaded on websites, social media accounts, or other digital information is one well-known way to improve accessibility for those with visual impairments. Furthermore, automatic picture captioning in surveillance systems can help interpret and analyze video by producing textual descriptions of the situations recorded. Furthermore, in the domain of content management and organization, these systems can help with effective visual content indexing and retrieval, which can lead to improved data management and searchability. Additionally, to improve the learning experience for students, educational platforms can incorporate image captioning technology to automatically generate descriptive captions for instructional images.

8. Conclusion

In our research paper here, we introduced an image description system that makes use of Convolution Neural Networks (CNNs) to extract features and Long Short-Term Memory (LSTM) networks to generate captions. Our model managed to obtain BLEU scores in comparison to conventional techniques, which suggests that it produced more precise and contextually appropriate descriptions. By incorporating attention mechanisms and adjusting on social media datasets we were able to improve the quality of the captions provided. However, despite these advancements challenges still remain such as handling various image contexts and the limitations associated with BLEU scores. In

the future research endeavors should involve more assessment criteria consideration of designs and enlargement of data sets to enhance model effectiveness further. Indeed we will focus on rectifying incomplete citations and grammatical problems, in the paper's ultimate rendition.

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